

EXPERIMENT – 2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

THEORY

Chunking is a cognitive technique that groups information into smaller, meaningful units. It helps users remember information more easily by reducing mental load.

In memory-based tasks, chunking allows users to recall groups of elements instead of individual items.

This improves retention, speed, and accuracy.

Using visual icons also strengthens memory through pattern recognition and association.

TOOLS USED

- Figma (UI Design Tool)
- Icons and Pixel Graphics
- Prototype Preview Tool
- System Browser

DESCRIPTION OF THE INTERFACE

The interface is designed as a sequence recall game with four main frames:

1. Instruction Page
2. Memorization Page
3. Recall Page
4. Result Page

Each frame guides the user step-by-step through the memory task.

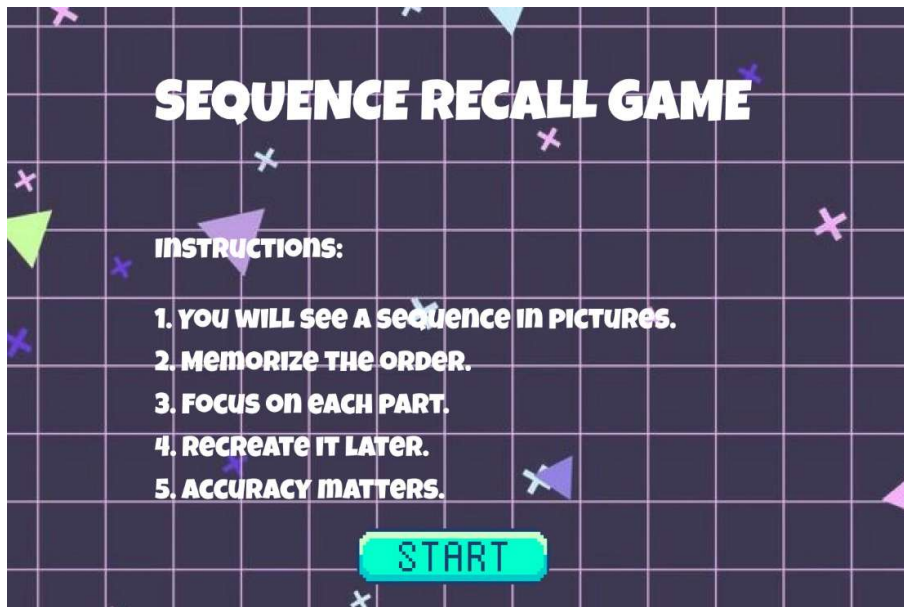
FRAME 1: INSTRUCTION PAGE

Description

This page displays instructions for the user before starting the task.

Chunking Analysis

1. Clear Numbering
 - Instructions are divided into steps.
 - Each step contains one action.
2. Logical Grouping
 - Steps 1–2: Observation
 - Step 3: Memorization
 - Steps 4–5: Recall
 - Step 6: Accuracy
3. Visual Hierarchy
 - Large title
 - Proper spacing
 - Highlighted Start button
4. Simplicity
 - Short sentences
 - Easy to understand language



Benefit

This reduces confusion and prepares users mentally for the task

FRAME 2: MEMORIZATION PHASE**Description**

Users view a sequence of images and memorize their order within a time limit.

Key Components

- Progress Bar
- Timer
- Grid of Icons
- Background Theme

Chunking Method

Users group icons based on:

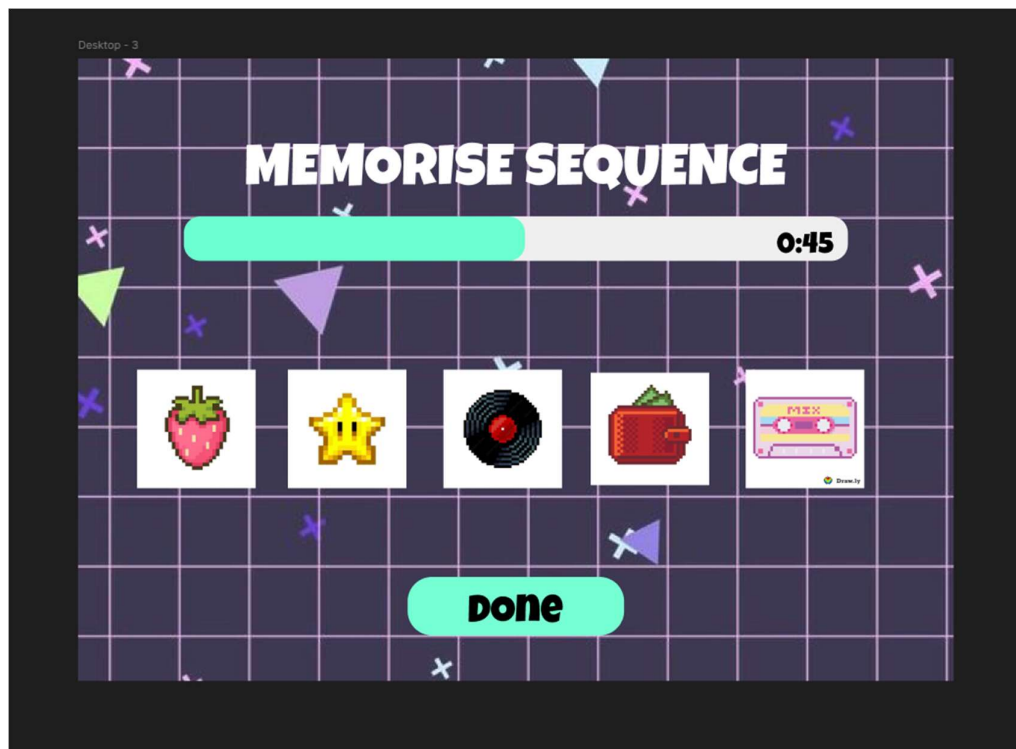
- Color
- Shape
- Category
- Position

Working

1. The user observes the images.
2. Timer creates urgency.
3. Users form mental groups.
4. Data is stored in short-term memory.

Benefit

- Improves visual memory
- Reduces overload
- Encourages pattern recognition



FRAME 3: RECALL PHASE

Description

Users select the correct sequence from multiple options.

Components

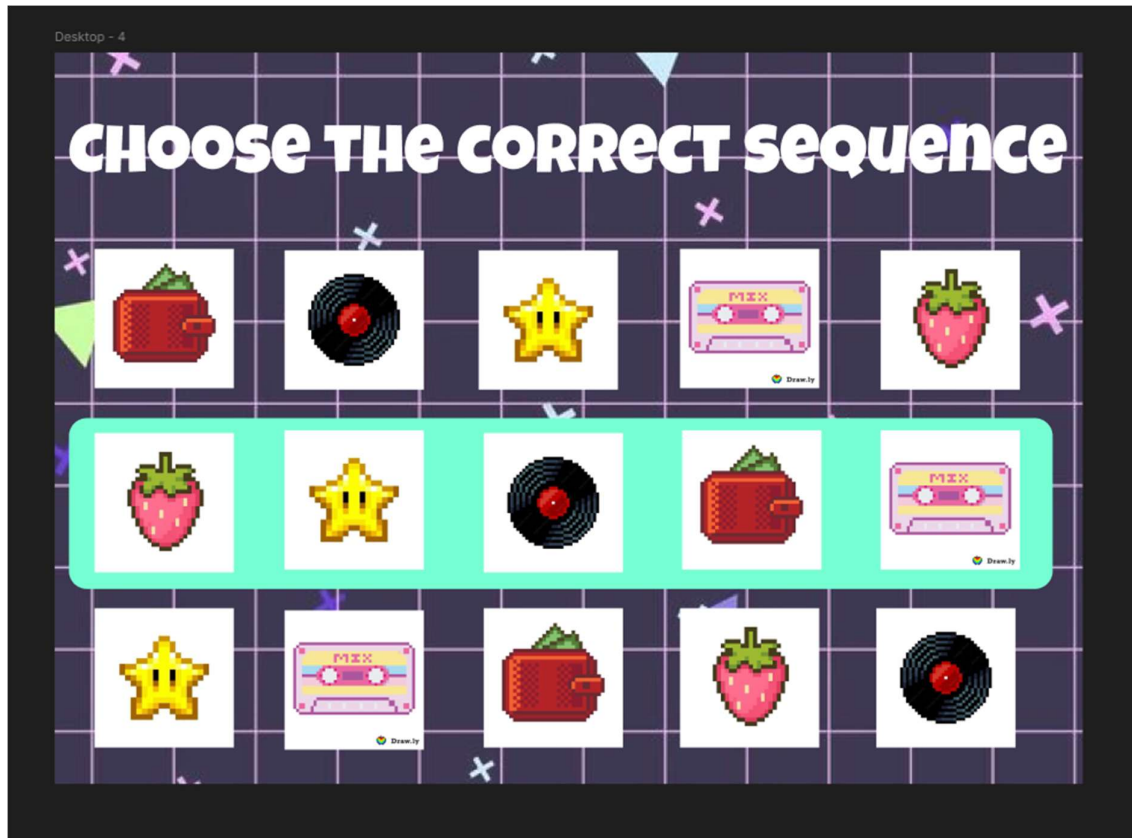
- Title
- Multiple Icon Choices
- Highlighted Selection Area
- Submit Button

Working

1. Users recall the memorized order.
2. They select icons in sequence.
3. Distractors test memory accuracy.
4. Selection is submitted.

Benefit

- Tests real recall
- Measures chunking effectiveness
- Improves decision-making



FRAME 4: RESULT PAGE

Description

Displays the final score after completion.

Components

- Score Display
- Feedback Message
- Play Again Button

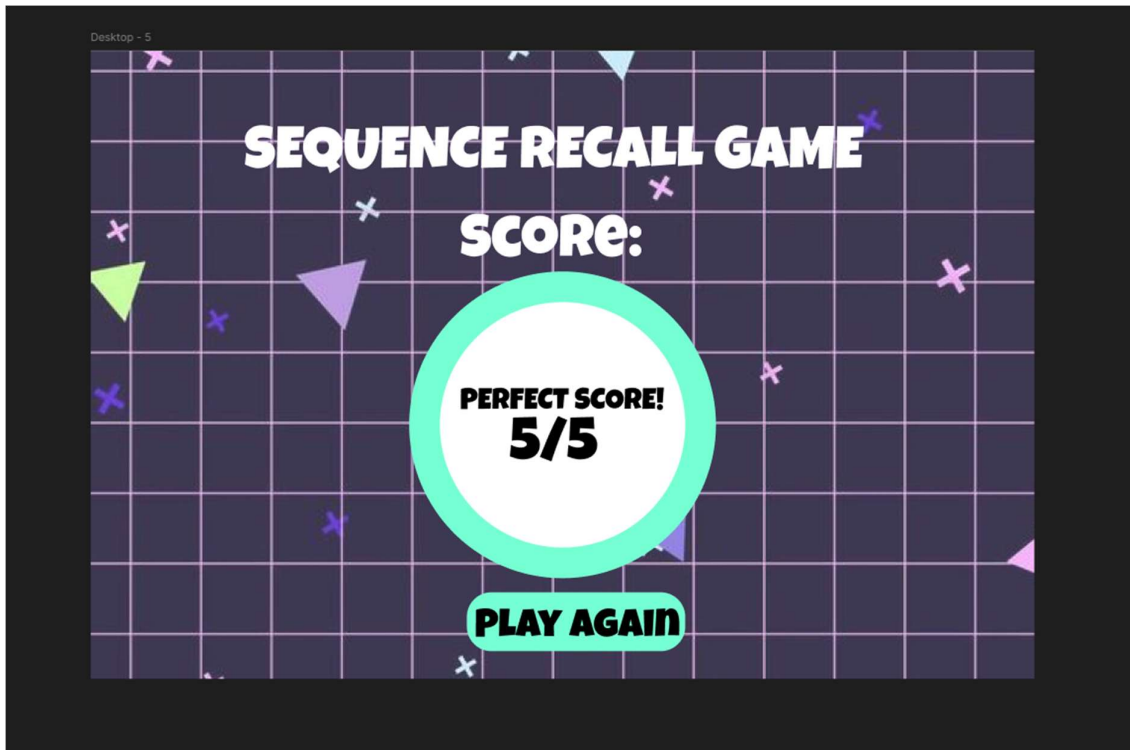
Working

1. System compares answers.
2. Score is calculated.

3. Result is displayed.
4. User may retry.

Benefit

- Motivates users
- Gives instant feedback
- Encourages improvement



EVALUATION AND ANALYSIS

The interface effectively applies chunking through:

- Grouped instructions
- Limited icons
- Repeated patterns
- Time-based learning

Users can easily form mental associations.

Performance improves when users use grouping strategies.

The gamified design keeps users engaged and focused.

RESULT

The experiment successfully demonstrates that chunking improves visual memory and recall accuracy.

Users were able to remember sequences efficiently and achieve high scores.

The UI was user-friendly and easy to navigate.

CONCLUSION

The visual memory recall interface effectively applies chunking principles.

It reduces cognitive load and improves recall performance.

This experiment proves that structured grouping enhances short-term memory.

The design is suitable for learning and training applications.

FUTURE SCOPE

- Add difficulty levels
- Increase sequence length
- Add performance tracking
- Include sound effects
- Introduce adaptive timing

REFERENCE

Prototype Link:

<https://www.figma.com/proto/RIJs7mDqzuMNgSbUG4GDyg/Untitled?node-id=0-1&t=XSRS9kNitIcccXeg-1>